

Indhuja Sivasamy

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EDUCATION

M.S. Computer Science

May 2023 - Dec 2024

University of Alabama, Huntsville; GPA : 4.0

Received a full scholarship during my internship period at NSSTC

Related Coursework: Data Structures and Algorithms, Database Systems, Cloud Computing, Advanced Operating System, Software Engineering Process

TECHNICAL SKILLS

Languages: HTML, CSS, JavaScript, Python, Java(DSA), Kotlin

Technologies: ReactJS, FastAPI (REST API), Docker, JSON, Kubernetes, AJAX

Database: SQL, MongoDB, Firestore

Web Tools: AWS(CloudFront, S3, EC2), Google Cloud Platform, Postman, Chrome DevTools

Other Tools: Android Studio, Jira, Slack, GitHub, Agile Methodology

WORK EXPERIENCE

Volunteer Software Developer, Bright Mind Enrichment & Schooling

Apr 2025 - Present

- Contributing to the Street Care Android app by developing and improving key features, including user signup forms and real-time updates synced with Firestore.
- Built the full signup form flow, from UI implementation using Kotlin to storing data in Firestore and automating integration with a shared Google Sheet using Firestore Cloud Functions.
- Collaborated with the data management team to align backend integration with their workflow and ensure accurate syncing.
- Participated in code reviews and supported onboarding of new joiners through documentation and knowledge sharing.

Software Intern, NSSTC(NASA-ITSC), Alabama

Aug 2023 - Aug 2024

- Developed a full-stack web application with React and Redux for the frontend and FastAPI for the backend, allowing real-time interactions and updates via REST APIs and WebSockets, deployed on AWS using EC2 and S3.
 - Designed and implemented a multi-tab interface using Material UI, optimizing UI/UX and improving metadata retrieval speeds by 30%, enabling efficient exploration and navigation through large environmental datasets.
 - Managed cloud infrastructure and application deployments using Docker Compose for environment consistency, reducing deployment times by 25%. Used Prettier to enforce consistent code formatting across the frontend codebase.
 - Enhanced the FCX website by integrating advanced 3D visualizations for five new instruments, focused on improving the representation of complex NASA datasets for significant scientific presentations.
 - Engineered a sophisticated ETL pipeline utilizing NumPy and Pandas, accelerating data transformation for Skew plots and 3D formats by 50%, with results stored efficiently in AWS S3.
 - Collaborated on diverse visualization strategies to enhance the FCX website, involving debugging and UI enhancements critical for improving data accessibility and usability during key events like SE-TIM and the AGU Fall Meeting.
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ACADEMIC PROJECTS

Car Scheduling System Project

Nov 2024

- Configured and deployed a car scheduling system using Minikube and Docker across multiple pods to encapsulate various segments of the system, promoting isolated development. Received extra academic credits for exemplary project execution.
- Managed Kubernetes for inter-pod communication, to coordinate the flow of car parts from storage to assembly and hence maintaining data consistency.

- Utilized a stack comprising JavaScript, PHP, and XMLHttpRequest running on an Apache Web Server to handle asynchronous operations via AJAX, achieving efficient client-server interactions without a database.

Cloud Computing Project

Nov 2023

- Developed the React frontend for a distributed voting application, focusing on user authentication and interaction, deployed on AWS EC2 with Docker containerization.
 - Managed backend and database integration using MongoDB and Docker Compose, ensuring efficient data management and seamless deployment.
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PERSONAL PROJECTS

Jan 2023 - May 2023

- Undertook personal projects to practice web development and cloud integration skills.
- Developed small-scale applications for expense tracking and food ordering.
- **Expense Tracker** : Built the app with React/Node.js, integrated AWS Lambda, API Gateway, S3, and EC2, created a Lambda function for expense logic, configured API Gateway for data submission, deployed server on EC2, and used S3 for static assets.
- **Food Order App** : Developed the app with React and Bootstrap, implemented sign-in/sign-up, enabled meal selection, and created a cart feature for order management.
- Successfully created applications that improved my technical skills and provided practical solutions for expense tracking and food ordering.